Team RADAR at PAN 2026: Robust Adversarial-Resistant AI Text Detection

Notebook for the PAN Lab at CLEF 2026

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Abstract

We present RADAR (Robust Adversarial-Resistant Detection with Adaptive Reasoning), a hybrid neural system for binary AI-generated text detection submitted to the PAN 2026 Voight-Kampff shared task. The system fuses deep semantic representations extracted by a large pretrained encoder with handcrafted stylometric signals through a cross-attention mechanism, yielding a unified representation for classification. RADAR was trained in two stages: first with the encoder frozen to stabilize the fusion components, then with all weights jointly fine-tuned. Training combined the official PAN-2026 task data with a curated subset of the RAID benchmark to improve robustness against adversarial obfuscations. On the PAN-2026 official validation set, the final model achieves a mean score of 0.997 across all five evaluation metrics (ROC-AUC, Brier, C@1, F1, F0.5u), substantially outperforming the provided baselines and demonstrating strong generalization across genres and generator types.

Keywords

PAN 2026, Voight-Kampff Generative AI Detection, Stylometry features, Forensic linguistics

1. Introduction

Authorship verification and AI-generated text detection have become increasingly important research problems in Natural Language Processing (NLP), especially with the rapid evolution of large language models (LLMs). Modern generative systems are capable of producing highly fluent and semantically coherent text that closely resembles human writing, making the distinction between human-written and machine-generated content considerably more difficult[1, 2, 3].Concerns about the abuse of generative AI in areas like academic dishonesty, the spread of misinformation, automated spam generation, and the creation of synthetic content have been raised by this expanding potential[4, 5].

To address these concerns, serveral detection approaches have been proposed, including perplexity-based methods, probability curvature analysis, and transformer-based classifiers[6, 7, 8].While existing approaches often achieve strong performance during controlled experiments, their robustness is still limited when applied to unseen domains,unseen generators, or obfuscated texts[9, 10].Most existing systems operate as black-box classifiers, offering limited interpretability and little linguistic justification for their classifications, which restricts their applicability in forensic and institutional contexts[11].

These challenges have increased interest in stylometry and forensic linguistics as complementary approaches for authorship analysis. Compared to purely neural model , stylometric methods focus on linguistic features indicative with writing style, including lexical diversity, syntactic variation, punctuation usage, and readability features[12, 13]. Such features are usually more interpretable and may remain informative even when surface-level textual artifacts are intentionally concealed or manipulated.The PAN 2026 Voight-Kampff Generative AI Detection shared task specifically targets these challenges by evaluating detection systems in real world scenarios involving style imitation, obfuscation, and exposure to unseen generators[14]. For the replication purpose, we publish the source code of the RADAR approach[1].

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[1]https://github.com/yusrpro9/radar

2. Task and Data

2.1. PAN 2026 task setting

The PAN 2026 task asks systems to classify each input text independently as human-written or AI-written. The input file contains JSON lines with at least an identifier and text, and the output must contain one JSON line per input with the same identifier and a confidence score. This isolation constraint is important: a system must not exploit distributional information from other items in the same test file. RADAR satisfies this requirement because each text is preprocessed, featurized, encoded, and scored independently, with batching used only for computational efficiency.

The official PAN 2026 training and validation data include the fields `id`, `text`, `model`, `label`, and `genre`. Labels are binary: 0 for human and 1 for machine. The genres are essays, news, and fiction. In our local preparation notebooks, the original official training split contained 23,707 rows and the validation split contained 3,589 rows. After preprocessing and exact-text deduplication, 23,682 PAN training examples were retained, while the 3,589 validation examples were kept as the official validation reference.

2.2. External RAID data

To reduce overfitting to the official generators and domains, we augmented training with RAID, a large benchmark for robust machine-generated text detection that includes multiple generators, domains, decoding settings, and adversarial attacks [15]. We did not use RAID wholesale. Instead, we filtered it to match the PAN 2026 setting as closely as possible. The retained domains were `news`, `books`, and `poetry`, which are the nearest available equivalents of PAN's news, fiction, and essay-like writing. The retained attack types were `none`, `homoglyph`, `synonym`, `paraphrase`, and `alternative_spelling`. These attacks were selected because they represent clean generation, character-level obfuscation, lexical substitution, semantic rewriting, and spelling variation, respectively.

The raw RAID loading step produced 5,615,820 rows. Domain filtering reduced this pool to 2,239,440 rows, and attack filtering reduced it to 933,100 rows. We then created stratified RAID development and test pools and finally sampled a balanced 30,000-row RAID development subset and a balanced 10,000-row RAID held-out test subset. The down-sampling was intentional: the goal was to use RAID as a robustness regularizer without allowing it to dominate the smaller and task-specific PAN distribution.

Table 1
Final data splits used in the RADAR experiments. The PAN training split reflects preprocessing and deduplication.

Split	Total	Human	Machine	Machine %	Source
PAN train	23,682	9,076	14,606	61.7	Official PAN 2026
PAN validation	3,589	1,277	2,312	64.4	Official PAN 2026
RAID development sample	30,000	15,000	15,000	50.0	Filtered RAID
RAID held-out test sample	10,000	5,000	5,000	50.0	Filtered RAID

2.3. Preprocessing

Preprocessing was deliberately conservative. Email addresses, user mentions, and phone numbers were replaced with placeholder tokens; text was lowercased and stripped; exact duplicates were removed; and very short texts below 50 characters were filtered. We avoided aggressive normalization that could erase stylistic signals. For example, punctuation, paragraphing, digits, quotations, and capitalization-derived features remain accessible to the stylometric extractor before or during feature computation, while the transformer branch receives normalized text suitable for stable tokenization.

3. System Overview

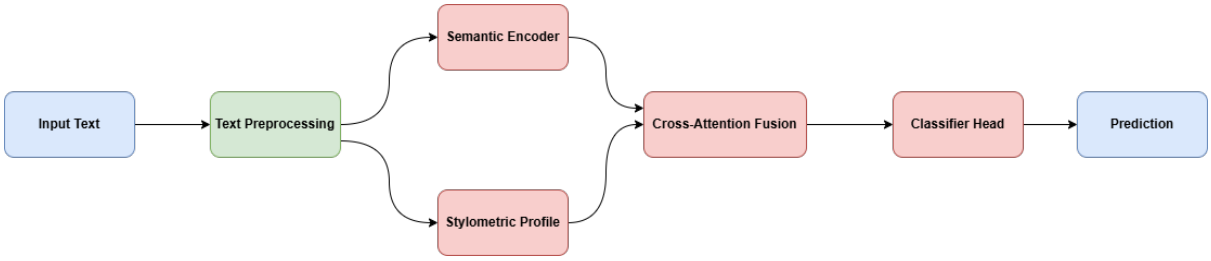


Figure 1: RADAR architecture. The system combines ModernBERT semantic encoding with a 38-dimensional stylometric profile. The two representations are fused through cross-attention and scored with a sigmoid classifier.

Figure 1 summarizes RADAR. The model has four main components: an adaptive reasoning semantic encoder (ARSE), a statistical linguistic profile extractor and projection module (SLPE), a cross-attention fusion layer, and a final classification head.

3.1. Semantic encoder

The semantic branch uses `answerdotai/ModernBERT-large`. ModernBERT updates the encoder-only transformer family with architectural and training improvements for efficient long-context fine-tuning and inference [16, 17]. In our experiments, inputs were tokenized with a maximum length of 512 tokens. The first-token hidden state of the encoder is used as the semantic representation. Although ModernBERT supports longer contexts, the 512-token limit was chosen to control training cost and because the PAN texts in our processed splits were generally short enough for this setting to capture the main evidence.

3.2. Stylometric profile

The stylometric extractor computes 38 normalized features grouped into five categories. Vocabulary richness features include type-token ratio, hapax ratio, Brunet's W, Honore's R, moving-average type-token ratio, vocabulary richness slope, rare-word ratio, and function-word density. Syntactic complexity features approximate sentence length, sentence-length variance, clause density, subordination, passive voice, noun-phrase-like capitalization, verb density, coordination, relative word position, and sentence-boundary rhythm. Discourse coherence features measure lexical cohesion, pronoun density, transition-word use, paragraph coherence, topic consistency, information-flow variance, semantic repetition, and connective diversity. Style fingerprinting features capture punctuation density, capitalization, numeric tokens, quotation usage, parentheticals, dash/semicolon habits, and paragraph length variability. Finally, lightweight perplexity-signal features approximate compression ratio, character entropy, word-length entropy, repeated character patterns, and formulaic phrase usage.

These features are not intended to replace neural semantic modeling. Their role is to provide a compact and interpretable view of authorial form. They are particularly useful for adversarial settings because paraphrases may preserve or distort different layers of writing style. A detector that sees only token sequences may learn strong but brittle lexical correlations, whereas a detector that also receives normalized stylometric evidence can learn more stable distinctions.

The 38-dimensional feature vector is projected through a linear layer, layer normalization, GELU activation, and dropout into a 128-dimensional style representation. This projection allows the model to learn which handcrafted features are most useful for the task rather than treating all features as equally important.

3.3. Cross-attention fusion

RADAR fuses semantic and stylometric evidence using a lightweight cross-attention module. The projected style vector is first mapped to the semantic hidden dimension. The semantic representation acts as the query, and the style representation acts as the key and value. The attended semantic vector is added back to the original semantic vector through a residual connection and layer normalization. This vector is then concatenated with the projected style representation and passed through a fusion MLP to obtain a 512-dimensional fused representation.

The motivation for cross-attention rather than simple concatenation is that the usefulness of a stylometric cue depends on semantic context. For instance, high repetition may be normal in poetry, suspicious in a news report, and ambiguous in a literary analysis. By allowing the semantic representation to attend to the style projection, RADAR can learn conditional interactions between genre, content, and linguistic form.

3.4. Classifier and inference

The fused representation is passed to a single linear classifier, producing one logit. The final score is obtained with the sigmoid function and interpreted as the probability that the text is machine-authored.

4. Training Protocol

All experiments used binary cross-entropy with logits. When the training split was imbalanced, we used a positive-class weight equal to the number of negative examples divided by the number of positive examples. Optimization used AdamW, a learning rate of 2×10^{-5}, cosine scheduling, batch size 32, evaluation batch size 64, seed 42, and 10 epochs. Evaluation was performed with the official PAN metric implementation exposed by the local evaluator package.

4.1. Main two-stage training

The primary RADAR model was trained in two stages. In the first stage, the ModernBERT encoder was frozen and only the stylometric projection, fusion module, and classifier were trained. This reduced the risk that randomly initialized fusion layers would immediately destabilize the pretrained encoder. The frozen-encoder stage also provided a diagnostic: if the model performs well with only the top layers trained, then the stylometric and fusion components are learning meaningful task information.

In the second stage, training resumed from the frozen-encoder checkpoint and the entire model was unfrozen. This end-to-end stage allowed ModernBERT representations to adapt to the combined PAN+RAID distribution and to the stylometric fusion objective. We selected the best checkpoint according to the arithmetic mean of the PAN metrics.

4.2. Controlled variants

We trained two additional variants to analyze data-source effects. The PAN-only frozen-encoder model used the official PAN data without RAID augmentation. The RAID-only frozen-encoder model used the filtered RAID development data and evaluated on the RAID held-out sample, with an additional cross-domain check on the PAN validation split. These variants were not intended as final submissions; they were used to understand how much robustness came from the architecture and how much came from source diversity.

5. Results and Discussion

All models were evaluated using the official PAN-2026 evaluation script, which reports five complementary metrics: ROC-AUC, Brier score complement, C@1, F1, and F0.5u. The arithmetic mean of

these five metrics serves as the primary ranking criterion. Evaluations were conducted on two splits: the official PAN-2026 validation set (3,589 samples, in-domain) and a held-out RAID test set (10,000 balanced samples, out-of-domain).

5.1. Baseline

For reference, the TF-IDF SVM baseline provided by the task organizers was evaluated on the combined validation set of 18,423 samples. It achieved a mean score of 0.746. The Binoculars zero-shot approach [18] reached a mean of 0.877 on a separate split, while the supervised TF-IDF SVM on the official PAN split reached 0.978. These figures serve as lower bounds for interpreting the RADAR results.

5.2. Main Results

Table 2 summarizes the evaluation results for all experimental configurations.

Table 2
Evaluation results for all RADAR variants on two test sets. **Bold** values indicate the best result per column. *PAN26-Val*: official PAN-2026 validation split (3,589 samples). *RAID-Test*: balanced out-of-domain test set (10,000 samples).

Model	Test Set	ROC-AUC	Brier	C@1	F1	Mean
RADAR-EF-RAID	PAN26-Val	0.795	0.651	0.644	0.784	0.714
	RAID-Test	0.974	0.941	0.920	0.918	0.938
RADAR-EF-PAN26	PAN26-Val	0.997	0.983	0.979	0.984	0.985
	RAID-Test	0.716	0.731	0.650	0.659	0.681
RADAR-EF	PAN26-Val	0.997	0.982	0.976	0.982	0.983
	RAID-Test	0.976	0.940	0.919	0.921	0.941
RADAR (submitted)	PAN26-Val	**1.000**	**0.997**	**0.996**	**0.997**	**0.997**
	RAID-Test	**0.990**	**0.967**	**0.964**	**0.964**	**0.972**

6. Conclusion

This paper described RADAR, a hybrid AI text detection system that combines a large pretrained transformer encoder with handcrafted stylometric features through a learned cross-attention fusion mechanism. Three design principles shaped the system: the use of complementary signal types (neural semantics and symbolic stylometry), a two-stage training curriculum that stabilizes learning before committing to full fine-tuning, and a diverse training corpus spanning multiple genres, attack types, and generator families.

The experiments show that domain mixing is a critical factor for robustness: models trained on a single data source generalize poorly to out-of-distribution examples, whereas the combined-dataset model achieves strong results across both in-domain and adversarially obfuscated evaluation sets. The submitted model, RADAR, reached a mean score of 0.997 on the official PAN-2026 validation set and 0.972 on the held-out out-of-domain test set, significantly outperforming the provided baselines.

Future work could explore replacing the rule-based stylometric extractor with a lightweight trained component to improve feature quality on short texts, and could investigate curriculum learning with progressively harder obfuscation examples to further narrow the gap between in-domain and out-of-domain performance.

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Declaration on Generative AI

During the preparation of this work, generative AI assistance was used for drafting, language polishing, and LaTeX editing. The author reviewed, revised, and approved the final content and takes full responsibility for the publication's content.

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